



Design and Test Report

by



TEAM VISHWA
10344

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Veermata Jijabai Technological Institute

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1 | Rover Mobility

1.1 | Mechanism Design

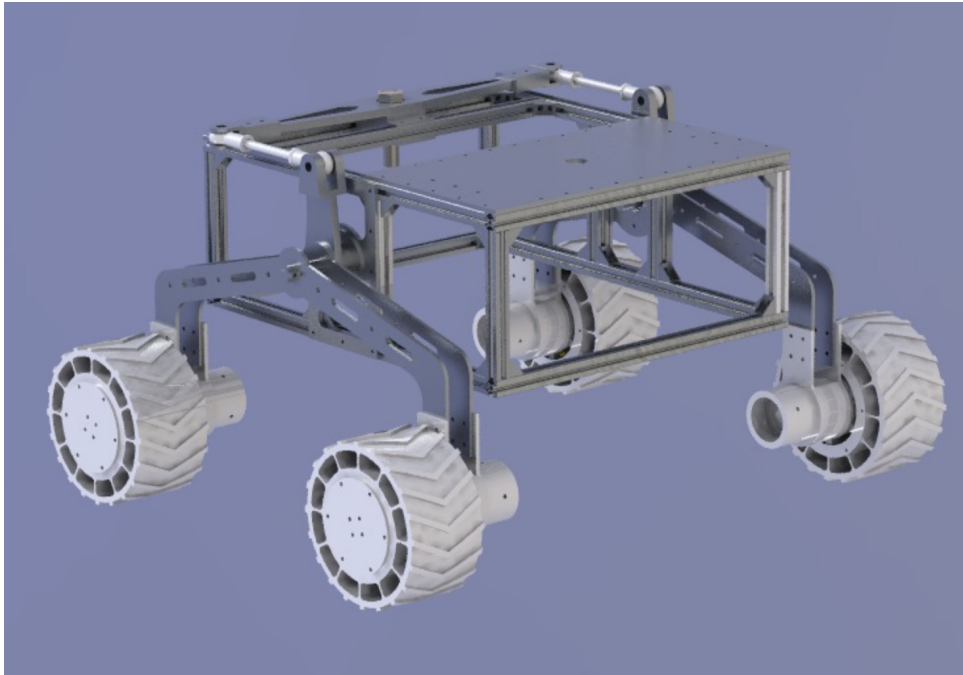


Figure 1.1: Rover alone

1. Chassis

- The rover's chassis is meticulously crafted using a robust box frame structure fashioned from Aluminium 6063-T6 Extruded Bars, each measuring 20 by 20 millimetres. These bars are skilfully interconnected through the strategic placement of corner brackets and sliding nuts, ensuring a cohesive and resilient framework.
- Through a thorough evaluation process, this approach emerged as the most optimal and practical solution for constructing the rover's basic structure. Various alternatives were considered during the decision-making process, with carbon fibre initially being among the contenders due to its exceptional strength-to-weight ratio. However, it was ultimately deemed economically unviable due to its high manufacturing costs. In contrast, aluminium-based materials presented a more financially feasible option while still offering adequate strength.
- Deliberations between solid aluminium bars and Aluminium 6063-T6 Extruded Bars (20 x 20) ensued, with the latter being selected for its ability to reduce weight without compromising structural integrity. To provide a tangible comparison, a 170- millimetre length of a solid aluminium bar weighs approximately 183 grams, while an extruded bar of identical dimensions weighs a mere 79 grams, indicating a nearly twofold reduction in weight. This significant disparity underscores the efficiency gained by opting for extruded bars. Moreover, the frame's robustness is further enhanced by strategically sheathing it with aluminium sheets in areas where high-load mechanisms such as the robotic arm and suspension will be situated.
- Conversely, in regions where the load is expected to be less substantial, acrylic sheets are employed, balancing strength requirements with weight considerations. This meticulous approach to material selection aims to optimize the rover's structural performance, ensuring it can withstand the rigors of its mission while maintaining an optimal balance between strength and weight.

2. Suspension

- The suspension system plays a critical role in ensuring the rover's ability to navigate uneven terrain with efficiency and stability. In our design, we have chosen Aluminium 6061 as the core metal for the various

linkages within the suspension system. Additionally, we've implemented protective casings to safeguard the more vulnerable components. Our suspension mechanism takes the form of an inverted V, where each side is connected with a differential. This configuration enables the rover to respond sensitively to uneven pitching on either side, ensuring continuous contact with the ground and maintaining the chassis parallel to the plane of motion.

- The links and differential components have been meticulously crafted using Aluminium 6061 sheets, chosen after careful consideration of multiple factors including strength, machinability, weight, and cost-effectiveness. For the wheel system, we've opted for a fully 3D printed approach, utilizing TPU (Thermoplastic Polyurethane) filaments to strike the optimal balance between strength and elasticity. This choice ensures that the wheels can withstand the rigors of varied terrain while maintaining flexibility.
- One of the significant challenges we encountered was determining the steering mechanism for the rover. After extensive analysis of available options, we settled on a differential speed control system. This system operates based on the variance in speed between the rover's drive motors. By adjusting the RPM (Revolutions Per Minute) of the motors on each side, we can induce a turning effect. For instance, if the motors on the right side run at a higher RPM, the rover will veer left, forming an arc around a point falling on the left side of the rover's frame.
- To ensure sufficient torque for propulsion, we've selected drive motors capable of delivering 140 kgf-cm torque. This choice was informed by considerations such as the rover's weight, the nature of the surface it will traverse, and the desired speed. To protect these motors and facilitate smooth rotation, we've designed custom printed wheel hubs containing integrated bearings. These hubs not only safeguard the motors but also contribute to the overall efficiency and durability of the rover's propulsion system.

1.2 | Drive Electronics design

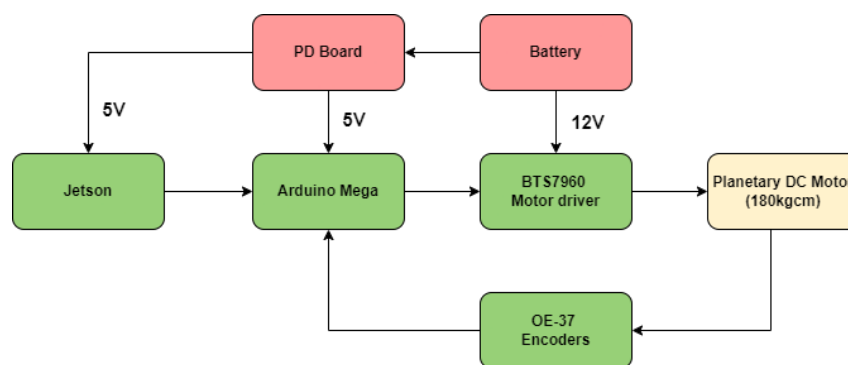


Figure 1.2: Drive System

- The rover's drive system prioritizes both robust starting torque for navigating challenging terrain and precise, consistent movements for controlled exploration.
- To achieve this balance, we've implemented a combination of powerful DC motors and a closed-loop control system.
- DC Motors with Planetary Gearboxes are chosen as they provide the necessary high torque needed to overcome obstacles and traverse uneven surfaces.
- Planetary gearboxes further amplify this torque, allowing the rover to handle demanding environments.
- Closed-Loop Control System ensures precise and consistent movements of the rover through continuous feedback.
- It comprises several key components:
 1. Custom BTS7960 Driver Boards: These boards control the motors based on input signals.
 2. OE-37 Magnetic Encoders: Mounted on the motors, these encoders provide real-time feedback on the rover's position and rotation.
 3. Arduino Mega Microcontroller: The brains of the operation, this microcontroller interprets encoder feedback, adjusts motor speed and direction using Pulse Width Modulation (PWM) signals, and ultimately corrects the rover's position for accurate movement.

1.3 | Design calculations

- The interconnected parts depicted in the provided image are assembled using T-slot sliding nuts and 5 mm diameter bolts, which are standardized (The only) for Aluminium 6063-T6 Extruded Bars measuring 20 x 20.
- This extends to the components mounted onto the frame, ensuring uniformity and structural integrity throughout. Through meticulous analysis, it was determined that this assembly configuration possesses the requisite strength and stability to support a load of up to 50 kg without the need for additional calculations or reinforcements.
- The arrow shows where the force acts on the flange which holds the shaft holding the suspension links.

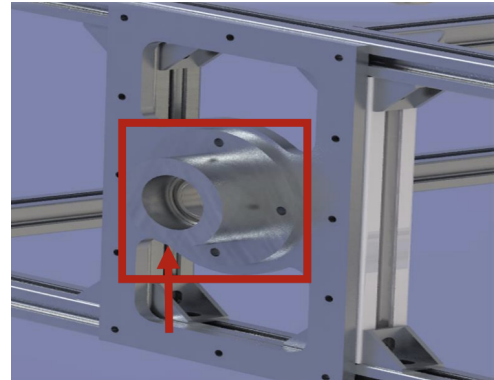
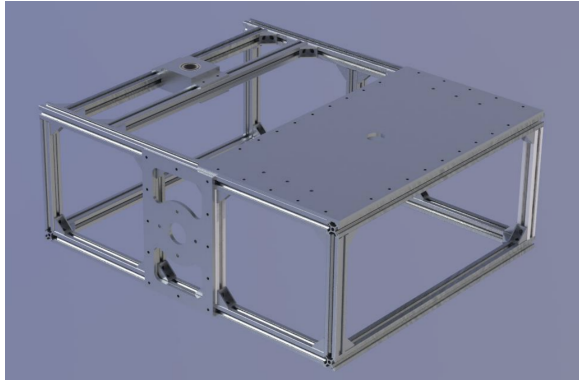


Figure 1.3: Chassis

- Design Calculation for Suspension flange

$$P = 50\text{kg} = 500\text{N}$$

Material taken is steel,

$$N = 4 \rightarrow (\text{No. of bolts})$$

$$\sigma_t = 140\text{N}/\text{m}^2$$

$$W = P/2$$

$$W = 500/2 = 250\text{N} \rightarrow (\text{Force acting on one flange})$$

- Primary load $\Rightarrow$ Direct shear

$$F_s = 250/4 = 62.5\text{N}$$

- Secondary load $\Rightarrow$ Tensile

$$F_t = (62.5 * 4 * 21 * 48) / 4 * 48^2 = 27.34375\text{N}$$

$$F_t - eq = 1/2[27.34375 + \sqrt{27.34375^2 + 4 * (62.5)^2}] = 77.64\text{N}$$

$$d_c^2 = 4 * 77.64 / 314 * 140 = 0.706$$

$$d_c = 0.84$$

$$d = d_c / 0.8 = 0.84 / 0.8$$

$$d = 1.05\text{mm}$$

Diameter $\approx 1\text{mm}$ but it was taken to be 5mm as a factor of safety.

1.4 | Software requirements and its design detail

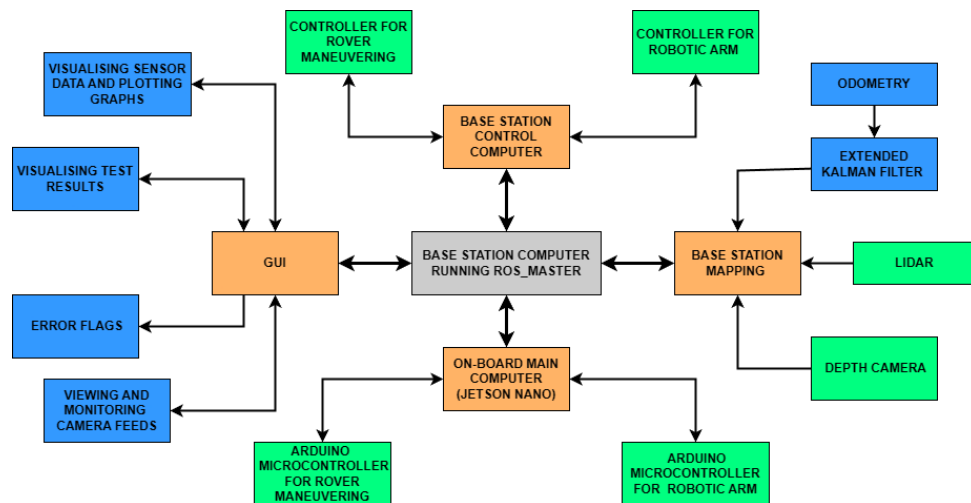


Figure 1.4: Rover Teleoperation Structure

- **ROS Packages:** ROS Installation on the Jetson. Chosen melodic based distribution taking into consideration the compatibility with Jetson and desired features. ROS packages relevant to rover control, navigation, sensor integration, and communication. This includes packages like `ros_control`, `navigation`, `sensor_msgs`, and custom packages for interfacing with specific sensors and actuators.
- **Sensor Drivers:** Developed ROS drivers for sensors such as encoders, IMUs, cameras, and LiDARs. These drivers enable ROS nodes to interface with hardware and receive sensor data. Taking sensor data from the sensor and sending it to the ROS interface through different communication protocols is also one of the ways to receive sensor readings in ROS distribution. These protocols include I2C, UART, etc.
- **Motor Control:** Implemented ROS nodes for controlling rover motors based on feedback from encoders and closed-loop control algorithms. Primarily the feedback mechanism code is being written in the controller level code with PID tuned wheel control. The ROS distribution only provides the target velocities and receives encoder feedback from the controller for closed loop mobility.
- **Communication Protocols:** Implemented ROS topics, services, and actions for inter-process communication within the rover software architecture. Here we have used standard ROS message types (e.g., `sensor_msgs`, `geometry_msgs`) for data exchange between different nodes.
- **User Interface:** A graphical user interface (GUI) using ROS visualization tools for monitoring rover status, visualizing sensor data, and sending navigation commands.
- **ROS Node Architecture:** Designed a modular ROS node architecture with nodes for sensor drivers, motor control, localization, mapping, path planning, and communication. Node Interconnection between ROS topics, services, and actions for data exchange between nodes. Also established communication pipelines for sensor data flow, control commands, and status updates.
- **Launch Files:** ROS launch files are being created to orchestrate the startup and configuration of multiple nodes and parameter settings. Using launch files simplifies deployment and reproducibility of rover software. Parameter Configuration and Definition are also included for tuning control algorithms, navigation parameters, sensor configurations, and system settings.



1.5 | Specifications of COTS components

1. Arduino Mega:

Microcontroller	ATmega2560
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
Digital I/O Pins	54(of which 15 provide PWM output)
Analog Pins	16
DC current per I/O Pin	20 mA
DC current for 3.3V Pin	50 mA
Flash Memory	256 KB of which 8 KB used by bootloader
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz
Length	101.52 mm
Width	53.3 mm
Weight	37 g

2. OE-37 Encoders:

Model	OE-37
Input Supply voltage (V)	4.5 ~ 24
Supply Current (mA)	14
Output Current (mA)	< 0.1
Output Leakage Current (μ A)	10
Operating relative humidity	20% ~ 85% RH
Operating Temperature ($^{\circ}$ C)	-40 ~ +85
Diameter (mm)	34.3
Weight (gm)	24



3. Jetson Nano:

Supply Voltage	5V 4A
GPU	128-core Maxwell
CPU	Quad-core ARM A57 @ 1.43 GHz
Memory	4 GB 64-bit LPDDR4 25.6 GB/s
Video Encoder	4K @ 30, 4x 1080p @ 30, 9x 720p @ 30 (H.264/H.265)
Video Decoder	4K @ 60, 2x 4K @ 30, 8x 1080p @ 30, 18x 720p @ 30, (H.264/H.265)
Camera	2x MIPI CSI-2 DPHY lanes
Connectivity	Gigabit Ethernet, M.2 Key E
Display	HDMI 2.0 and eDP 1.4
Length (mm)	100
Width (mm):	80
Height (mm):	29

4. LiFePo4 batteries:

Nominal Voltage /V	12V
Nominal Capacity /Ah (0.2C)	≈12Ah
Operating Voltage Range	10V-14.6V
Design	4S2P
Cell type/ Chemistry	32650/ LFP
Weight	1210 gms
Dimensions (Lx W x H)	130 x 73 x 67 mm
Charging Connector	XT-60 Female Connector
Discharging Connector	XT-60 Female Connector
Max. Continues Charge Current	6A
Charging Duration	≈120 mins
Max Voltage	14.6V
Min Voltage	10V
Max. Continues Discharge Current	12.0A

2 | Navigation and Identification

2.1 | Software requirements and design details

- ROS Melodic distribution for compatibility with Jetson Nano. Also packages like Navigation stack for path planning and localization like move_base, amcl. Implement Adaptive Monte Carlo Localization (AMCL) algorithm for accurate localization of the rover within the environment.
- Sensor_msgs package for sensor data communication. Use depth camera data to generate a 3D map of the environment for obstacle detection and terrain analysis.
- Depth camera drivers compatible with ROS Custom ROS packages for implementing specific functionalities like obstacle avoidance, object detection. Depth Camera compatible with ROS distribution.
- Install the required ROS drivers to interface it with ROS nodes.
- Object Detection Models that include pre-trained learning models for object detection.
- Tensorflow or Pytorch frameworks for training custom object detection with dimension specifications.
- These also include the addition of Libraries like OpenCV library for image processing and computer vision tasks, Point cloud processing libraries like PCL - Point Cloud Library for processing depth data.
- Utilize sensor fusion techniques to combine data from multiple sensors for robust obstacle detection.
- Localization and Mapping: Utilize ROS navigation stack for localization and mapping using SLAM to enable the rover to understand its position and environment.
- Path Planning and Navigation: Develop ROS nodes for path planning algorithms with A* search and navigation controllers to generate and execute trajectories while avoiding obstacles. Utilize ROS navigation stack for global and local path planning.



Figure 2.1: Obtained 2D Occupancy Grid

3 | Manipulator

3.1 | Mechanism Design

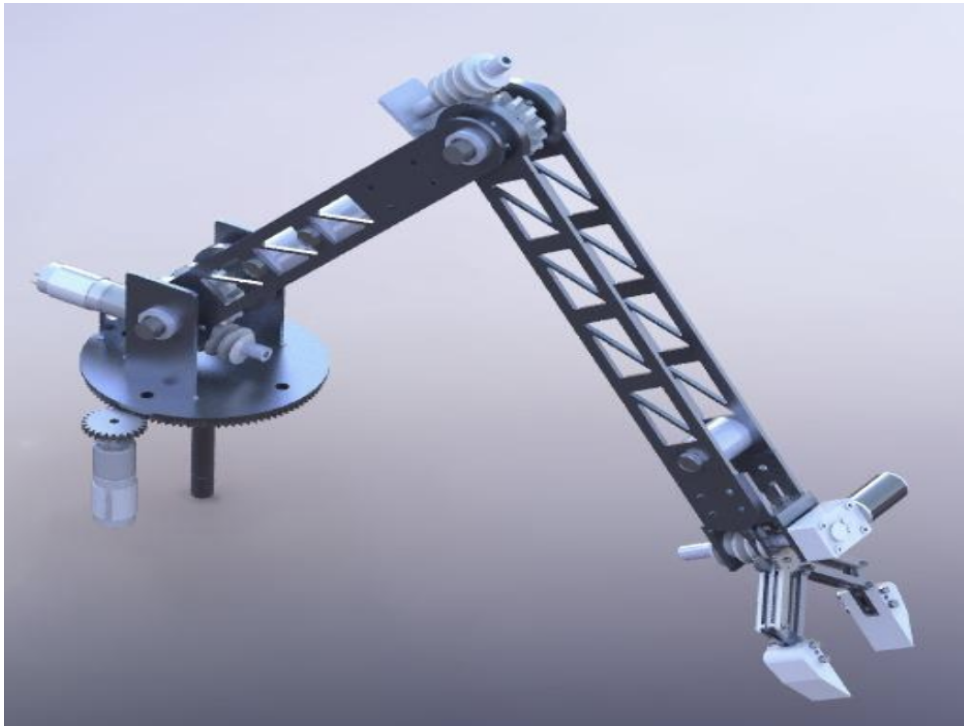


Figure 3.1: Manipulator

- We've integrated a sophisticated 3 Degree of Freedom (DOF) robotic arm into our rover, designed to handle a variety of tasks with precision and endurance. This arm boasts dynamic capabilities in kinematics, allowing for effective motion across a wide range of scenarios.
- Its movements are orchestrated through inverse kinematics, ensuring precise positioning for tasks such as picking and placing payloads.
- Crafted from Al6061 sheets, the arm strikes an optimal balance between strength and weight, crucial for its performance in space-constrained environments.
- To bolster its capabilities, we've incorporated planetary gear motors with worm gears, featuring a 15:1 reduction ratio. This configuration significantly enhances torque at the elbow and shoulder joints, enabling a diverse range of angular motions while minimizing backlash, ensuring smooth operation even in challenging conditions.
- The base of the arm utilizes a robust spur gear system, powered by a planetary gear motor, delivering ample torque for rotational movements.
- Moreover, a support system has been implemented to reinforce the base shaft, preventing any unwanted cantilevering of the arm during operation.
- Through strategic weight optimization, specific regions of the arm have been targeted to reduce overall weight without sacrificing functionality, enhancing the rover's efficiency and versatility in its tasks.
- This gripper features multiple articulated fingers that can open and close to grasp objects with precision.
- The design seems to allow for both fine control and a strong grip, enabling the robotic arm to pick up and manipulate objects of varying shapes and sizes.

3.2 | Manipulator Electronics design

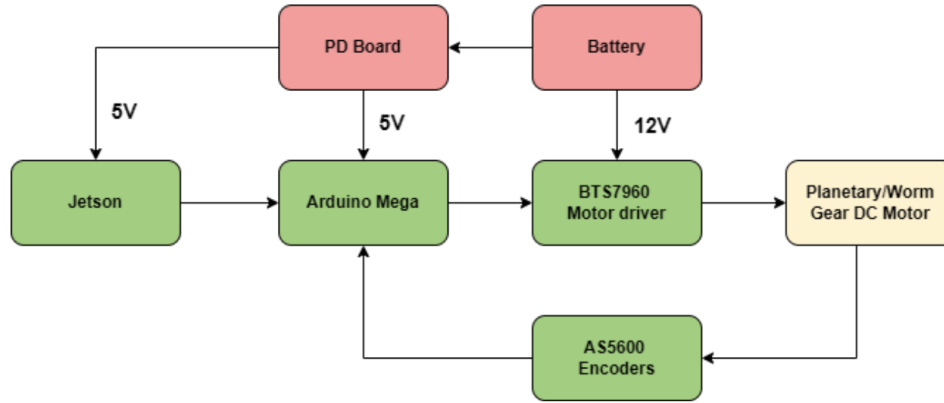


Figure 3.2: Manipulator System

- The rover employs 3 DOF arm consisting of three joints allowing movement in three different directions. Each joint provides rotational movement around its axis, allowing the arm to reach various positions in 3D space.
- DC motors are used along with AS5600 encoders which work in a closed loop system which ensures that the arm's position accurately matches with the desired position. The Arduino Mega microcontroller processes input commands and position feedback from the encoders, adjusting the motor speeds via the BTS7960 motor driver to achieve the desired positions.
- The AS5600 is a magnetic rotary position sensor that detects the angular position of a magnet. It operates by measuring changes in the magnetic field and offers high precision, typically with resolutions down to 12 bits, providing accurate position feedback for precise movement of the manipulator.

3.3 | Design calculations

The Arm calculations were performed for a theoretical load of 18Kg at a distance of 1m.

Torque at 1m length of arm

$$= 18 \times 9.8 \times 1 = 176.4Nm$$

To produce such torque, a planetary gear motor was used with a torque of 12Nm. This was coupled with a 15:1 ratio worm gear, which had a bronze wheel & steel worm, to reduce friction losses. Generally, the efficiency of worm gears lies in the range of 50-70%. Since our system had sufficient lubrication & the advantage of different ratios, the efficiency was found to be around 60-62%. Thus we consider,

$$\eta_{gear} = 0.6.$$

Assuming motor efficiency to be 90% due to windage & miscellaneous losses, we have,

$$\eta_{motor} = 0.9$$

$$\eta_o = \eta_{gear} \times \eta_{motor} = 0.6 \times 0.9 = 0.54$$

$$\text{Output Torque} = \text{Motor Torque} \times \text{Gear Ratio} = 12 \times 15 = 180Nm$$

$$\text{Actual Torque} = 180 \times \eta_o = 97.2Nm$$

The worm is practically able to lift a payload of 8-9 Kgs, considering inertial factors as well.

The competition requirements dictate a target load of 5kgs. Hence, the system is dynamically capable of lifting the payload.

Bolt Calculations: For bolts used in Links and Support Plates (6 bolts??).

For a load of 5kg, $F = 5 \times 9.8 = 49N$

$$\text{Force per bolt} = 49/6 = 8.167N$$

Primary Stress - Direct Shear

$$\tau_p = 8.167/s$$

Secondary Stress - Torsional Shear

$$F = 49 * 1000 * 60 / 6 * 60^2 = 136.11N(\text{Same for all bolts})$$

$$\tau_s = 136.11/s$$

$$\tau_r = \sqrt{\tau_p^2 + \tau_s^2}$$

$$= 136.4/sN/m^2$$

For a mild steel bolt with $N = 2$,

$$\sigma_y = 280N/mm^2$$

$$\sigma[T] = 0.5 \times 140 = 70N/mm^2$$

$$\therefore 70 = 136.4/s$$

$$d_c^2 = 2.48 \implies d_c = 1.575mm \implies M2$$

We have used M8 Bolts.

$\therefore$ FOS is increased 16 times to account for variable or fatigue loading. The bolted joints are safe.

3.4 | Software requirements and its design detail

- **ROS Installation on the Jetson:** Install ROS Melodic on the Jetson platform, ensuring compatibility and desired features for manipulator control. Utilize ROS packages relevant to manipulator control, kinematics, sensor integration, and communication. This includes packages such as `ros_control`, `moveit`, `sensor_msgs`, and custom packages for interfacing with specific sensors and motors.
- **Sensor Drivers:** Develop ROS drivers for sensors including encoders, IMUs, cameras, and any other sensors required for manipulator control. These drivers enable ROS nodes to interface with hardware and receive sensor data through communication protocols such as I2C, UART, etc.
- **Manipulator Control:** Implement ROS nodes for controlling manipulator joints based on feedback from encoders and closed-loop control algorithms. Develop feedback mechanism code within the controller nodes, incorporating PID-tuned joint control. The ROS distribution provides target joint angles or velocities and receives encoder feedback for closed-loop manipulation.
- **Communication Protocols:** Implement ROS topics, services, and actions for inter-process communication within the manipulator software architecture. Utilize standard ROS message types (e.g., `sensor_msgs`, `geometry_msgs`) for data exchange between different nodes.
- **ROS Node Architecture:** Design a modular ROS node architecture with nodes for sensor drivers, manipulator control, kinematics, trajectory planning, and communication. Establish communication pipelines for sensor data flow, control commands, and status updates between nodes.
- **Launch Files:** Create ROS launch files to orchestrate the startup and configuration of multiple nodes and parameter settings. Utilize launch files for parameter configuration and definition to tune control algorithms, kinematics parameters, sensor configurations, and system settings.
- **Real-Time Performance:** Optimize ROS software for real-time performance on the Jetson platform, implementing custom optimizations to reduce latency and improve responsiveness for manipulator control. Designed test cases to verify functionality, compatibility, and performance under various manipulation scenarios.



3.5 | Specifications of COTS components

1. Arduino Mega:

Microcontroller	ATmega2560
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
Digital I/O Pins	54(of which 15 provide PWM output)
Analog Pins	16
DC current per I/O Pin	20 mA
DC current for 3.3V Pin	50 mA
Flash Memory	256 KB of which 8 KB used by bootloader
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz
Length	101.52 mm
Width	53.3 mm
Weight	37 g

2. OE-37 Encoders:

Model	OE-37
Input Supply voltage (V)	4.5 ~ 24
Supply Current (mA)	14
Output Current (mA)	< 0.1
Output Leakage Current (μ A)	10
Operating relative humidity	20% ~ 85% RH
Operating Temperature ($^{\circ}$ C)	-40 ~ +85
Diameter (mm)	34.3
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3. Jetson Nano:

Supply Voltage	5V 4A
GPU	128-core Maxwell
CPU	Quad-core ARM A57 @ 1.43 GHz
Memory	4 GB 64-bit LPDDR4 25.6 GB/s
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Video Decoder	4K @ 60, 2x 4K @ 30, 8x 1080p @ 30, 18x 720p @ 30, (H.264/H.265)
Camera	2x MIPI CSI-2 DPHY lanes
Connectivity	Gigabit Ethernet, M.2 Key E
Display	HDMI 2.0 and eDP 1.4
Length (mm)	100
Width (mm):	80
Height (mm):	29

4. LiFePo4 batteries:

Nominal Voltage /V	12V
Nominal Capacity /Ah (0.2C)	≈12Ah
Operating Voltage Range	10V-14.6V
Design	4S2P
Cell type/ Chemistry	32650/ LFP
Weight	1210 gms
Dimensions (Lx W x H)	130 x 73 x 67 mm
Charging Connector	XT-60 Female Connector
Discharging Connector	XT-60 Female Connector
Max. Continues Charge Current	6A
Charging Duration	≈120 mins
Max Voltage	14.6V
Min Voltage	10V
Max. Continues Discharge Current	12.0A

4 | Communications

4.1 | RF design

Our communication system consists of an omnidirectional antenna aboard the rover and a sector antenna at the base station. Both antennas are powered using their compatible PoE injectors. Jetson Nano is used as a central single board computer to relay sensor data and camera feed to the base station via the antenna. Multiple IP Camera feed is relayed to the Jetson Nano using a PoE Switch, which further transmits it to the base station. Specifications: Two Rocket M5s are been used for point-to-point (P2P) and point-to-multipoint (P2M) communication. It features incredible range performance (50+km) and breakthrough speed (150+Mbps real TCPI/IP).

Communication Standards:

- 5 GHz frequency band (5.450-5.850 GHz): This band will be utilized for video transmission from various cameras on the rover, sensor data relay and joystick commands from the base station.
- This frequency range is commonly used for wireless networking and provides advantages such as less interference and higher data rates compared to the 2.4 GHz band.
- Gain of Base Station antenna: 19dBi Gain of the antenna at the rover: 13dBi

4.2 | Software requirements and design details

Device	Model	Location	Frequency Band
Ubiquiti airMax Omnidirectional Antenna	M5 – 13dBi AMO-5G13	Mounted on the rover	5.450-5.850 GHz
Ubiquiti airMax Sector Antenna (120-degree sector angle)	M5 – 19dBi AM-5G19-120	One at the Base Station	4.900-5.850 GHz
Ubiquiti Networks High Power Base MIMO Station	Rocket M5	One on the rover with Omnidirectional antenna One on the base station with Sector antenna	5.150-5.850 GHz



4.3 | Specifications of COTS components

1. Rocket R5AC-lite:

Dimensions	162 x 84 x 37 mm	
Weight	250 g	
Power Supply	24V, 0.5A Gigabit PoE Adapter	
Power Method	Passive PoE (Pairs 4, 5+; 7, 8 Return)	
Max. Power Consumption	8.5W	
Operating Frequency	5150 - 5875 MHz	
Processor	Atheros MIPS 74Kc, 720 MHz	
Memory	128 MB DDR2 SDRAM, 16 MB NOR FLASH	
Networking Interface	(1) 10/100/1000 Mbps	
RF Connections	(2) RP-SMA (Waterproof	
LEDs	Power, LAN, (4) Signal Strength	
Channel Sizes	PtP Mode	PtMP Mode
	10/20/30/40/50/60/80 MHz	10/20/30/40 MHz
Enclosure Characteristics	Outdoor UV Stabilized Plastic	
Supported Voltage Range	20-26VDC	
ESD/EMP Protection	± 24KV Air / Contact	
Operating Temperature	-40 to 80° C (-40 to 176° F)	
Operating Humidity	5 to 95% Non Condensing	
RoHS Compliance	Yes	
Security	WPA2 AES Only	
QoS	Supports Packet Level Classification WMM and User Customer Level: High/Medium/Low	
Statistical Reporting	Up Time, Packet Errors, Data Rates, Wireless Distance, Ethernet Link Rate	



2. OMNI-Directional Antenna:

Model	AMO-5G13
Dimensions	799 x 90 x 65 mm
Weight	0.82 kg
Frequency Range	5.45 - 5.85 GHz
Gain	13 dB
Elevation Beamwidth	7°
Max VSWR	1.5:1
Downtilt	2°
Wind Survivability	200 km/h
Wind Loading	84.52 N @ 200 km/h
Polarization	Dual-Linear
Cross-pol Isolation	25 dB min
ETSI Specification	EN 302 326 DN2
Mounting	Universal Pole Mount, RocketM Bracket, and Weatherproof RF Jumpers Included

3. Sector-Antenna:

Model	AM-5G19-120
Dimensions	700 x 135 x 73 mm
Weight	5.9 kg
Frequency Range	5.15 - 5.85 GHz
Gain	18.6 - 19.1 dBi
HPOL Beamwidth	123° (6 dB)
VPOL Beamwidth	123° (6 dB)
Electrical Beamwidth	4°
Electrical Downtilt	2°
Max. VSWR	1.5:1
Wind Survivability	200 km/h
Wind Loading	137.9 N @ 200 km/h
Polarization	Dual-Linear
Cross-pol Isolation	28 dB Min
ETSI Specification	EN 302 326 DN2
Mounting	Universal Pole Mount, RocketM Bracket, and Weatherproof RF Jumpers Included

5 | Emergency response system

5.1 | Kill switch design

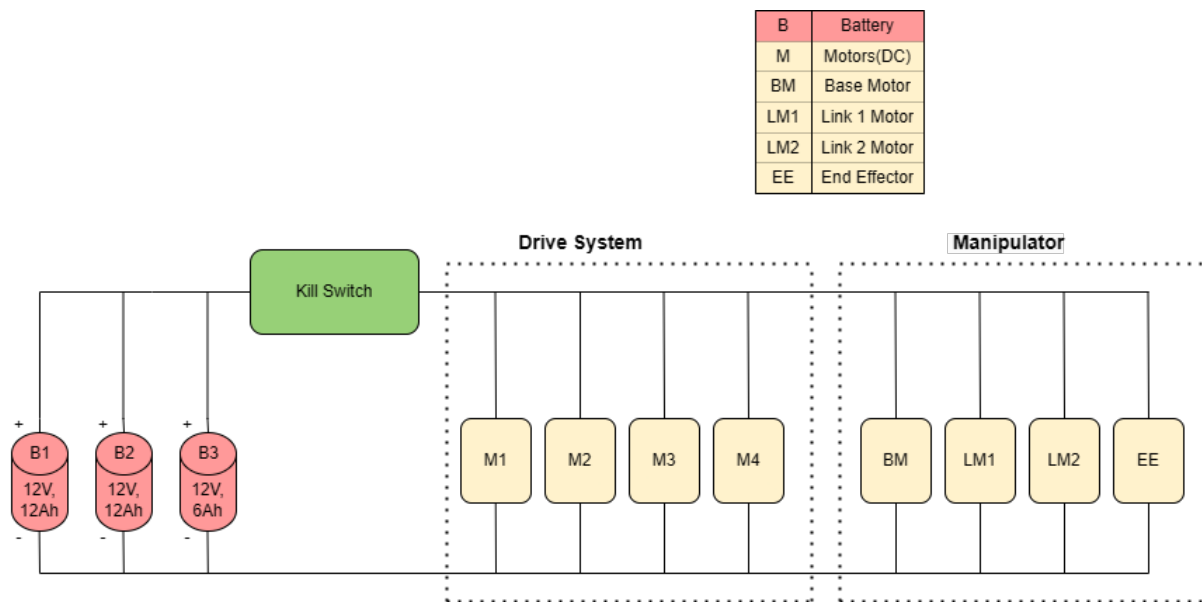


Figure 5.1: Kill Switch design

The rover is powered by 3 LiFePO4 Batteries connected in parallel which yields a total capacity of 30Ah power supply at 12V. A Kill Switch is connected in series with these batteries and the output load which are the motors responsible for propulsion and movement of the robotic manipulator. When the Kill switch is disengaged, it acts as a closed switch thereby, completing the circuit and power is supplied to the various motors of the rover. When the Kill switch is in engaged state, it acts as an open switch, cutting off the power supply to the various motors of the rover. The Kill switch is employed to immediately stop all operations of the drive and manipulator system, helping to prevent accidents, damage or undesirable outcomes.

■ Specifications of the Kill Switch used:

Rated Insulation Voltage:	600V
Agreed Thermal Current :	10A
Switch Top Diameter:	39 mm
Switch Mount Neck Diameter:	23 mm
Material:	Plastic and metal
Weight (g):	50

NOTE: Specifications taken from Robu and may not match with the specifications of our kill switch

6 | Analysis

Analysis	Force/Load	Location
on suspension flange	400 Newton	given on the whole of flange
on the sheet where plant is going to be placed	400 Newton	given on the whole of the plate
on the robotic arm base	100000 Nmm	at the holes where the shaft will be placed
on aluminium extrusion bar i.e 410 mm in length	50000 Nmm	on the center of the aluminium extrusion bar

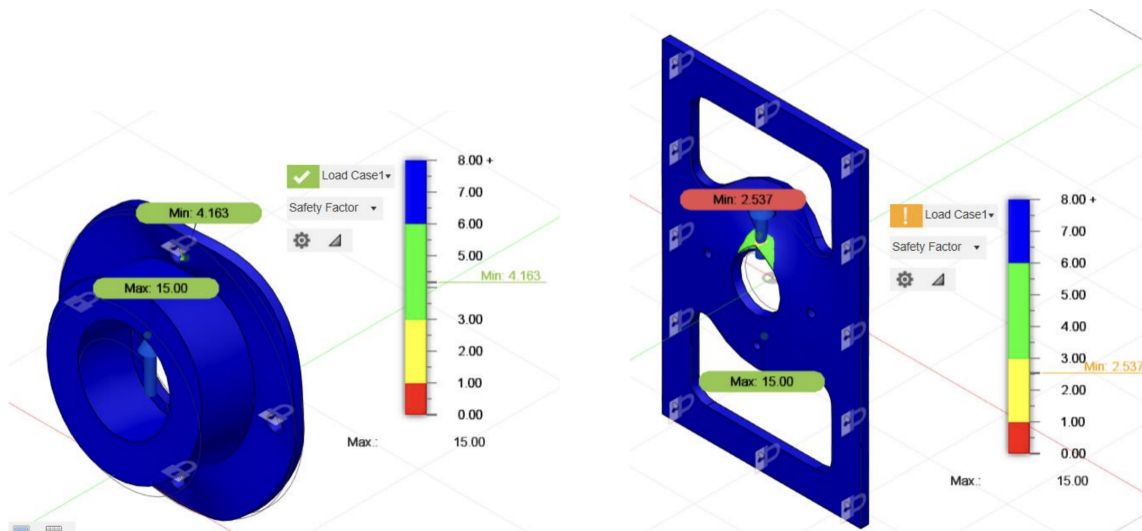


Figure 6.1: Flange and Sheet

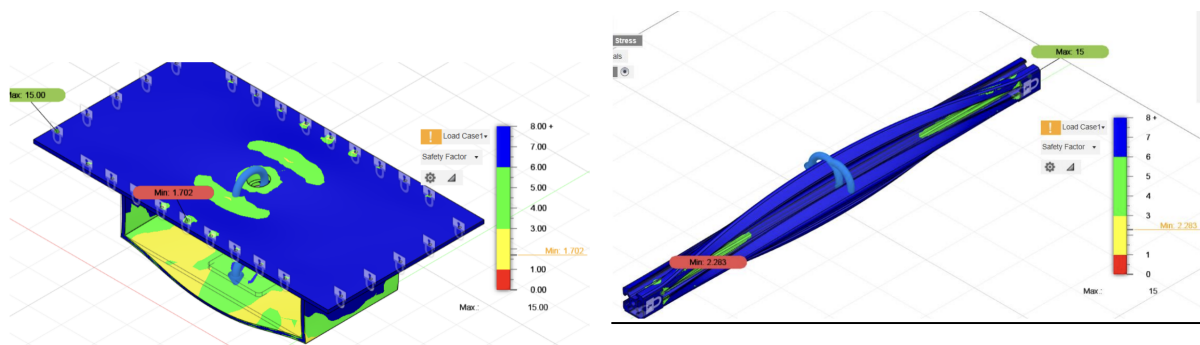


Figure 6.2: Arm Base and Extrusion bar

7 | Weighing and dimensional measurement details of the rover

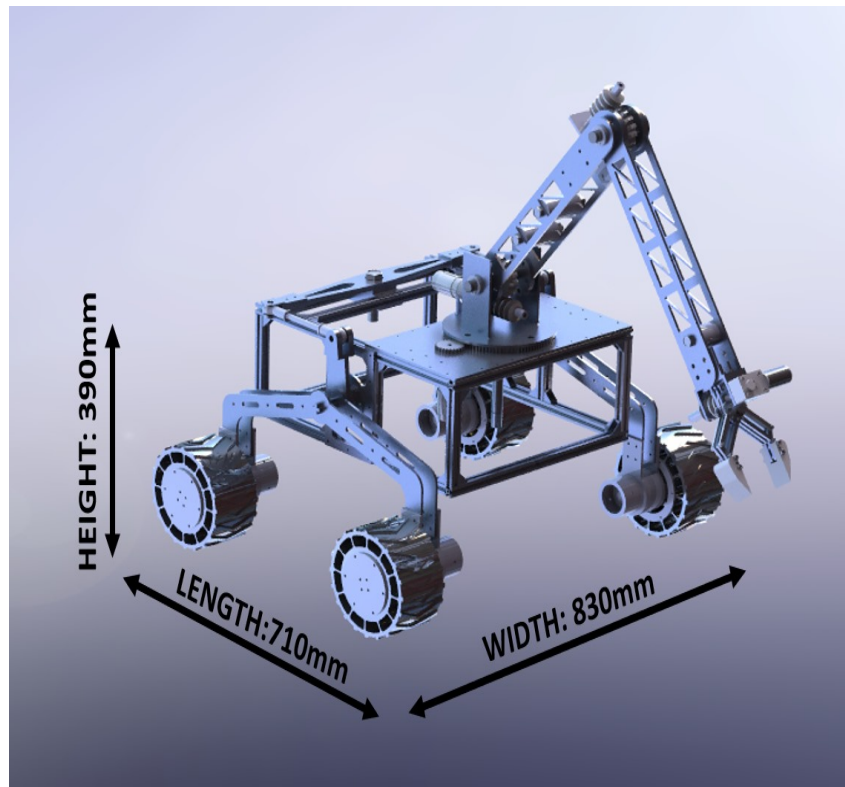


Figure 7.1: The above figure shows the dimensions of the rover



Figure 7.2: The above figure shows the Weight of the rover